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# OS-Tutor-RAG

A retrieval-augmented, Socratic tutor for the Operating Systems course, with a measured evaluation of its pipeline

CC: [COURSE CODE] CN: [COURSE NAME]

**Adhyan Jain (24BCI0111) · Atharva Sheersh Pandey (24BCI0069)**

Faculty: [FACULTY NAME] · October 2026

# Introduction & Objective

Why a grounded chatbot is not yet a tutor

## The problem

- **Slides are passive.** Terse decks and dense chapters never explain; they assume you know what to look for.
- **General chatbots are ungrounded.** Fluent, but a student cannot tell an explanation from a plausible invention.
- **Plain RAG is still a search box.** It answers what was asked and gives a beginner no way in.

### Our idea

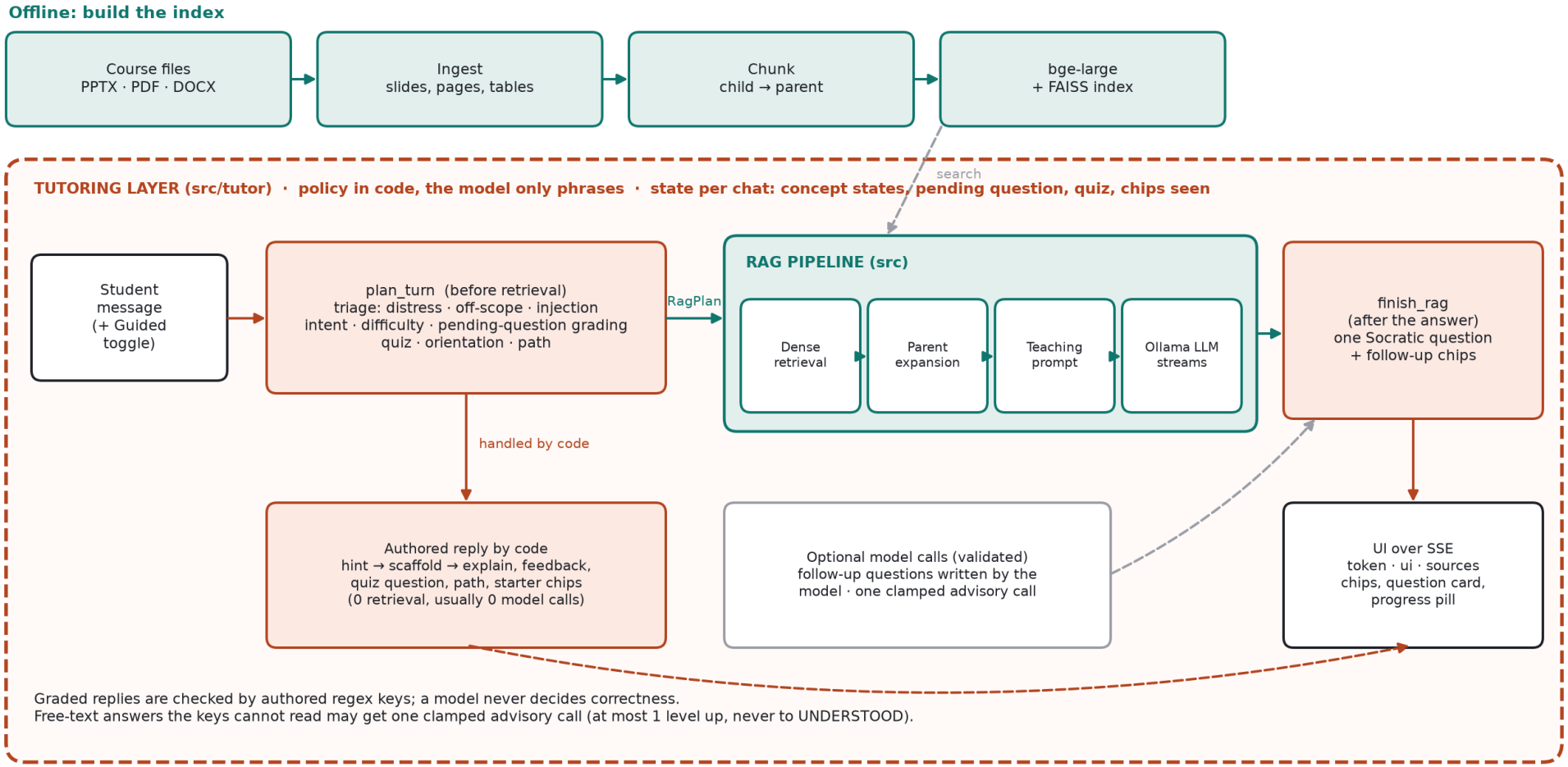
Keep every answer grounded in the course files, and let tested code, not the model, decide how to teach.

## Objectives

- **O1** Index PPTX, PDF and DOCX without silently losing content.
- **O2** Compare chunking, retrieval, reranking and prompts on a 26-question golden set; adopt only what measures better.
- **O3** Measure the evaluation's own noise so small differences are not over-read.
- **O4** Streaming chat: ELI5 / undergrad / exam-prep levels, cited sources.
- **O5** Socratic tutor: 30 concepts, 96 graded questions, hint → scaffold → explain, 16 misconception probes, quiz mode, kind handling of struggle and distress.
- **O6** Orientation, learning path, starter and follow-up chips for a student who knows nothing.

# Methodology

The tutoring layer wraps the RAG pipeline on every turn




**Small-to-big retrieval (match small chunks, read the parent slide) · policy in code (state, grading, next question) · the model only phrases**

# Results & Discussion (1/2): Prompt and RAG findings

Measurement overturned the plan: fixing defects and the prompt beat choosing techniques

## Scored prompt ("few_shot")

```
Answer the question using only the context
below. If the context doesn't contain the
answer, say so.

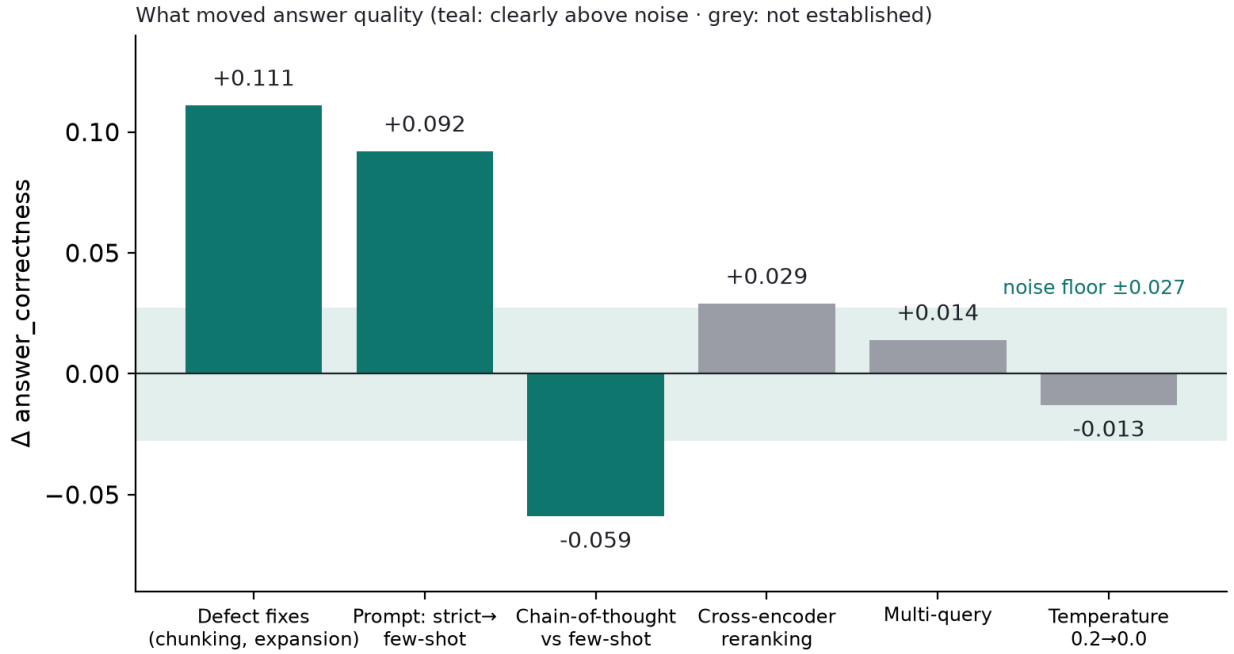
Answer in continuous prose, stating each fact
once. Do not add preamble about the context.

Example question: What are user mode and
kernel mode?
Example answer: CPU hardware has multiple
privilege levels: user mode runs user code,
and kernel mode runs OS code ...

Context: {context}  Question: {query}
```

Scoring: RAGAS, judge qwen2.5:7b (a different model family from the llama3 generator). Per question the judge sees question, retrieved contexts, answer and ground truth; answer_correctness is an F1 over statements. 26 questions, 5 topics.

**The app's teaching prompt (explain first; the tutor appends the question) is a different prompt and was not RAGAS-scored.**




### Defects

Titles indexed as chunks: refusals 11.7% → ~0% once fixed.

### Prompt shape

+0.092 few_shot; CoT −0.059 (answers 27% shorter).

### Noise ±0.027

Same config scored 0.7752 and 0.7478. Dense R@5 = 1.000.

# Results & Discussion (2/2): The tutor in action

Real engine output for a new student, and how the tutoring layer was checked

I'm new to operating systems

A good way in is to build up from the foundations. Here is where I'd start: Operating system, User mode and kernel mode, System calls and traps.

[model's grounded answer]  Think about this: Why do we put an OS between programs and the hardware, instead of letting each program talk to the hardware directly?

no idea

That's completely fine, you don't need to know this already. ... An easier way in: Imagine ten programs all writing to the same disk at once. What would you want in the middle ...

It hides the hardware behind a simple abstraction, and lets many programs share it safely without ...

Exactly. The OS gives programs a simple abstraction of the ...  [chips: explain with an example · test my understanding · how does it connect to ...]

## What the engine does

30 concepts · 96 graded questions · 16 misconceptions. State: UNKNOWN → INTRODUCED → ATTEMPTED → PARTIAL → UNDERSTOOD → MASTERED. Hint → scaffold → explain after 2 misses. Quiz: 3 questions, easy to hard.

## How it was checked

877 tutor tests + 427 existing pass: every model answer passes its own key, state never regresses, advisory call clamped, triage never blocks a course question, scripted new-student dialogue, API event contract. Live run via the web UI with llama3.1:8b and mistral.

## Honest limits

No student study and no RAGAS-style metric for pedagogy. Regex keys can miss a paraphrase (the clamped advisory call helps, not fixes). Local 7B models. The tutor is verified, not yet shown to improve learning.

# Conclusion

### What we built

A grounded, streaming OS tutor over the course files that explains first, then asks one graded Socratic question, gives hints, probes misconceptions, runs a quiz, helps a student who is stuck or knows nothing, and suggests good follow-ups.

### What we learned

Silent defects and the prompt shaped quality far more than retrieval technique. The benchmark's own noise (±0.027) must be measured before trusting differences. Policy in tested code made the tutor checkable.

### Limits and next

26 questions, 7B judge, single runs; corpus grew after measuring; teaching prompt not RAGAS-scored; no learner study. Next: re-measure on 36 files with repeats, validate the judge, pre/post study with students, paraphrase-robust grading.

### Code, evidence and findings log

**github.com/adhyan-jain/OS-Tutor-RAG (FINDINGS.md · paper/PAPER_FINAL.md for the RCR study)**

# References


**[1]** Lewis et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 33. arXiv:2005.11401

**[2]** Karpukhin et al. (2020). Dense passage retrieval for open-domain question answering. EMNLP 2020. arXiv:2004.04906

**[3]** Gao et al. (2022). Precise zero-shot dense retrieval without relevance labels (HyDE). arXiv:2212.10496

**[4]** Es et al. (2023). RAGAS: Automated evaluation of retrieval augmented generation. arXiv:2309.15217

**[5]** Zheng et al. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. arXiv:2306.05685

**[6]** Xiao et al. (2023). C-Pack: Packed Resources For General Chinese Embeddings (BGE). arXiv:2309.07597

**[7]** Wood, Bruner & Ross (1976). The role of tutoring in problem solving. J. Child Psychol. Psychiatry 17(2), 89-100

**[8]** Arpaci-Dusseau & Arpaci-Dusseau. Operating Systems: Three Easy Pieces (course reading)

**[9]** OS-Tutor-RAG repository: github.com/adhyan-jain/OS-Tutor-RAG


arXiv identifiers [1]-[6] checked against arxiv.org on 2026-10-24; [7] and [8] are standard journal/textbook citations.